

16 - Naive Bayes Classifier. Fisher's Argument for LDA. Evaluating a Classifier: AER, APER, Confusion Matrix, Precision, Recall, Specificity, F-measure. Cross-Validation: LOO and k -Fold.

Applied Statistics

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April 9–10, 2026

1 Big Picture

- We are in the **classification** chapter. We observe a point $\underline{x} \in \mathbb{R}^p$ and want to assign it to one of g groups via a rule:

$$\delta : \mathbb{R}^p \rightarrow \{1, 2, \dots, g\}$$

- Previously we built **QDA** and **LDA** by assuming each group is Gaussian and inverting the sample covariance matrix.
- This lecture addresses three follow-up questions:
 1. What if we *cannot* invert the covariance? $\rightarrow$ **Naive Bayes**.
 2. How do we *measure* how good a classifier is? $\rightarrow$ **AER, APER, cross-validation, confusion matrix metrics**.
 3. Can we derive LDA *without* assuming Gaussianity? $\rightarrow$ **Fisher's argument**.

2 Naive Bayes Classifier

2.1 The Problem with QDA/LDA

- QDA requires computing S_k^{-1} for each group k .
- If $p > n_k - 1$, the sample covariance S_k is **singular** (not invertible): the deviation vectors live in an at most $(n_k - 1)$ -dimensional subspace and cannot span $\mathbb{R}^p$.
- Sparse features (missing data) make this even worse.
- **Fix:** assume a **parametric model** for Σ_k to reduce the number of parameters.

2.2 Independence Assumption

- The strongest (and most common) simplification: assume the components of $\underline{x}$ are **conditionally independent** given the group.
- This means Σ_k is diagonal:

$$\Sigma_k = \text{diag}\left(\sigma_{11}^{(k)}, \dots, \sigma_{pp}^{(k)}\right)$$

- A diagonal matrix is trivially invertible and its determinant is the product of its diagonal entries.

2.3 The Naive Bayes (Gaussian) Discriminant Function

- Plugging the diagonal Σ_k into the QDA discriminant gives, for $k = 1, \dots, g$:

$$d_k^Q(\underline{x}) = \log(p_k) - \frac{1}{2} \log(\det \Sigma_k) - \frac{1}{2} \sum_{j=1}^p \frac{(x_j - \mu_{kj})^2}{\sigma_{jj}^{(k)}}$$

where μ_{kj} is the j -th component of $\underline{\mu}_k$.

- **Decision rule:** assign $\underline{x}$ to the group $k^* = \arg \min_k d_k^Q(\underline{x})$ (equivalently, the group maximising the score).

2.4 Estimation

- Estimate μ_{kj} with the sample mean of feature j in group k : $\hat{\mu}_{kj} = \bar{X}_{kj}$.
- Estimate the variance of feature j in group k :

$$\hat{\sigma}_{jj}^{(k)} = \frac{1}{n_k - 1} \sum_{\substack{i=1 \\ \ell_i=k}}^n (x_{ij} - \bar{X}_{kj})^2$$

- **Note:** this requires at least **2 observations per group per feature**.
- **Note:** this is doable even when $p \gg n$.

2.5 Why “Naive”?

- Assuming features are independent given the class is almost always wrong in practice (e.g. pixels, gene expressions, and word counts are all correlated).
- Despite this, the classifier often works surprisingly well.
- **Intuition:** to classify correctly you only need the *ranking* of posterior probabilities to be right, not the exact values. Biased but approximately right probabilities often give the correct ranking.

2.6 Generalisations

- Replace the Gaussian with any univariate distribution $\rightarrow$ **Naive Bayes (Poisson)**, **Naive Bayes (Bernoulli)**, etc.
- Can further assume equal variances across groups: $\sigma_{jj}^{(k)} = \sigma_{jj}$ for all k (Naive Bayes analogue of LDA's pooled covariance).
- **Warning:** doing **PCA first** and then classifying is a **bad recipe**. PCA finds directions of maximum overall variance, which need not separate classes. Fisher's method (Section 4) does this correctly.

3 Evaluating a Classifier

3.1 Actual Error Rate (AER)

- **Definition:** the probability that a randomly drawn unit from the population is misclassified:

$$AER(\delta) = \sum_{k=1}^g \sum_{i \neq k} \int_{R_k} f_i(\underline{x}) p_i d\underline{x}$$

where R_k is the region of $\mathbb{R}^p$ assigned to class k , f_i is the density of group i , and p_i its prior probability.

- **Problem:** f_i and p_i are unknown $\rightarrow$ AER cannot be computed directly; it must be estimated.

3.2 Confusion Matrix

- Apply δ to the training set. For $g = 2$ groups:

		Predicted label	
		L_1	L_2
2*Actual label	L_1	n_{11}	n_{12}
	L_2	n_{21}	n_{22}

- Diagonal entries = correct classifications; off-diagonal = errors.
- $n_{11} + n_{12} = n_1$, $n_{21} + n_{22} = n_2$, $n_1 + n_2 = n$.

3.3 Apparent Error Rate (APER)

- Count mistakes on the *same* training set used to build δ :

$$APER(\delta) = \frac{n_{12} + n_{21}}{n}$$

- Decomposition that reveals its structure:

$$APER(\delta) = \frac{n_1}{n} \cdot \frac{n_{12}}{n_1} + \frac{n_2}{n} \cdot \frac{n_{21}}{n_2} = \hat{p}_1 \int_{R_2} \hat{f}_1(\underline{x}) d\underline{x} + \hat{p}_2 \int_{R_1} \hat{f}_2(\underline{x}) d\underline{x}$$

- **Why APER is bad:**

- **Optimistic bias:** δ was built to minimise error on this very data $\rightarrow$ APER systematically underestimates the true error.
- **Prior issue:** $\hat{p}_i = n_i/n$ is only a valid estimate of p_i if the training set is a random sample from the population (often not the case).

4 Cross-Validation

4.1 Motivation

- We need to test the classifier on data it has **not seen during training**.
- Ideal: a separate held-out test set $\tilde{\mathbb{X}}$. Then:

$$\widehat{AER}(\delta) = \frac{1}{N} \sum_{i=1}^N \epsilon_i, \quad \epsilon_i = \begin{cases} 1 & \text{if } \tilde{\ell}_i \neq \hat{\ell}_i \\ 0 & \text{otherwise} \end{cases}$$

- **Note:** always split data **randomly** – never take the first k rows, as they may all belong to the same group.
- In practice we have only one dataset $\rightarrow$ simulate test sets via cross-validation.

4.2 Leave-One-Out (LOO / L1O)

- For $i = 1, \dots, n$ repeat:
 1. Remove unit i : $\mathbb{X}_{-i} = [x_1^T \ell_1, \dots, x_n^T \ell_n]^T$ without row i .
 2. Train δ_{-i} on $\mathbb{X}_{-i}$.
 3. Predict: $\hat{\ell}_i = \delta_{-i}(\underline{x}_i)$.
 4. Record: $\epsilon_i = \mathbf{1}\{\hat{\ell}_i \neq \ell_i\}$.

- Estimate AER:

$$\widehat{AER}_{LOO}(\delta) = \frac{1}{n} \sum_{i=1}^n \epsilon_i$$

- **Note:** $\widehat{AER}_{LOO}$ is an estimate of the AER for δ trained on the *full* dataset.
- **Note:** **LOO** = **n -fold cross-validation**.
- **Problems:** computationally expensive (n classifiers); **high variance** (explained below).

4.3 k -Fold Cross-Validation

- **Algorithm:**
 1. Choose $k < n$ (typically $k = 5$ or 10). **Randomly permute** the rows of $\mathbb{X}$, then split into k parts.
 2. For $j = 1, \dots, k$:

- (a) Hold out part j ; build $\mathbb{X}_{-\text{part}_j}$.
- (b) Train $\delta_{-\text{part}_j}$ on $\mathbb{X}_{-\text{part}_j}$.
- (c) Apply to part j and compute error rate:

$$Err_j = \frac{1}{n_j} \sum_{i \in \text{part}_j} \epsilon_i, \quad \epsilon_i = \begin{cases} 1 & \text{if } \delta_{-\text{part}_j}(x_i) \neq \ell_i \\ 0 & \text{otherwise} \end{cases}$$

- 3. Combine (weighted average):

$$\widehat{AER}(\delta) = \frac{1}{n} \sum_{j=1}^k n_j Err_j$$

- **Note:** the average may be *weighted* because parts need not be exactly equal in size.
- **Note:** if $k = n$ we recover LOO.

4.4 Why k -Fold Reduces Variance

- **LOO problem:** the n classifiers $\delta_{-1}, \dots, \delta_{-n}$ are trained on nearly identical datasets (differ in only 1 row) $\rightarrow$ errors ϵ_i are **strongly correlated**.
- Averaging correlated quantities does *not* reduce variance:

$$\text{Var}\left(\frac{1}{n} \sum_i \epsilon_i\right) \approx \text{Var}(\epsilon_j) \quad (\text{correlated case})$$

- **k -fold:** holding out k observations at a time $\rightarrow$ training sets are more different $\rightarrow$ classifiers less correlated $\rightarrow$ averaging genuinely reduces variance:

$$\text{Var}\left(\frac{1}{n} \sum_{j=1}^k n_j Err_j\right) < \text{Var}(Err_j)$$

- **Bias-variance trade-off:** smaller $k \rightarrow$ lower variance but higher bias (training on less data). LOO: minimum bias, maximum variance.
- **Key insight:** averaging reduces variance *only* when the averaged quantities are (approximately) independent.

4.5 Repeated k -Fold and Confidence Intervals

- Repeat k -fold B times with independent random permutations $\rightarrow$ obtain $\widehat{AER}_1(\delta), \dots, \widehat{AER}_B(\delta)$.
- Mean estimate:

$$\widehat{AER}_m(\delta) = \frac{1}{B} \sum_{b=1}^B \widehat{AER}_b(\delta)$$

- Variance estimate:

$$\widehat{\text{Var}}(\widehat{AER}(\delta)) = \frac{1}{B-1} \sum_{b=1}^B \left(\widehat{AER}_b(\delta) - \widehat{AER}_m(\delta) \right)^2$$

- $(1 - \alpha)$ confidence interval (CLT, choose B large):

$$CI_{1-\alpha} = \left[\widehat{AER}_m(\delta) \pm z_{\alpha/2} \sqrt{\widehat{\text{Var}}(\widehat{AER}(\delta)) \cdot \frac{1}{B}} \right]$$

- **Note:** B can be chosen up to $n!$ (the number of distinct row permutations).
- **Note:** this variability estimate is **impossible with LOO** – permuting the rows of a LOO procedure gives the same result every time.
- **Conclusion:** cross-validation lets us select the **best model** among many candidates.

5 Confusion Matrix Metrics

- For $g = 2$ (class 1 = positive, class 0 = negative):

		Predicted		Row total
		1	0	
2* Actual	1	n_{11}	n_{10}	$n_{1\cdot}$
	0	n_{01}	n_{00}	$n_{0\cdot}$
Col total		$n_{\cdot 1}$	$n_{\cdot 0}$	n

- **Accuracy:**

$$\text{Accuracy} = \frac{n_{11} + n_{00}}{n} = 1 - \text{APER}(\delta)$$

- **Precision** (Positive Predictive Value, PPV): of all units predicted positive, how many truly are?

$$\text{Precision} = \frac{n_{11}}{n_{\cdot 1}} = 1 - \text{FDR}$$

- **Recall** (Sensitivity, Power, True Positive Rate): of all truly positive units, how many did we catch?

$$\text{Recall} = \frac{n_{11}}{n_{1\cdot}} = 1 - P(\text{Type II error})$$

- **Specificity** (True Negative Rate): of all truly negative units, how many did we correctly label negative?

$$\text{Specificity} = \frac{n_{00}}{n_{0\cdot}} = 1 - P(\text{Type I error}) = 1 - \alpha$$

- **F-Measure** (harmonic mean of Precision and Recall):

$$F = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

- **Why so many metrics?** Accuracy alone is misleading on imbalanced data. Example: if 1% of patients are sick, a classifier that always says “healthy” achieves 99% accuracy but Recall = 0 and $F = 0$.
- Use **Precision** when false alarms are costly (e.g. cancer diagnosis); use **Recall** when missing positives is dangerous (e.g. infectious disease screening); use **F** for a balanced summary.

6 Fisher’s Argument for LDA

6.1 Motivation

- LDA was derived under the Gaussian assumption. Fisher shows the *same classifier* can be derived **without** any distributional assumption $\rightarrow$ LDA is **robust** to violations of Gaussianity.
- Fisher’s view also gives a **supervised dimensionality reduction** method: far better than PCA for classification purposes.
- **PCA vs. Fisher:** PCA finds the direction of maximum *total* variance – this may be exactly the direction where classes overlap. Fisher finds the direction of maximum *class separation*.

6.2 Setup

- Assume $\underline{X} \mid L = i \sim (\underline{\mu}_i, \Sigma)$ for $i = 1, \dots, g$. Same covariance Σ for all groups. **No Gaussian assumption required.**
- Pick any direction $\underline{a} \in \mathbb{R}^p$ and project:

$$\mathbb{E}[\underline{a}^T \underline{X} \mid L = i] = \underline{a}^T \underline{\mu}_i, \quad \text{Var}[\underline{a}^T \underline{X} \mid L = i] = \underline{a}^T \Sigma \underline{a}$$

6.3 Between- and Within-Group Variability

- **Between-group covariance matrix:**

$$B = \frac{1}{g-1} \sum_{i=1}^g (\underline{\mu}_i - \bar{\underline{\mu}})(\underline{\mu}_i - \bar{\underline{\mu}})^T, \quad \bar{\underline{\mu}} = \frac{1}{g} \sum_{i=1}^g \underline{\mu}_i$$

- **Within-group covariance matrix:** Σ (common to all groups).

6.4 The Optimisation Problem

- **Goal:** find $\underline{a}$ that maximises the ratio of between-group to within-group variability:

$$\arg \min_{\underline{a} \in \mathbb{R}^p} \frac{\underline{a}^T B \underline{a}}{\underline{a}^T \Sigma \underline{a}} = \arg \min_{\underline{a} \in \mathbb{R}^p} \frac{1}{g-1} \sum_{i=1}^g (\underline{a}^T \underline{\mu}_i - \underline{a}^T \bar{\underline{\mu}})^2 \cdot \frac{1}{\underline{a}^T \Sigma \underline{a}}$$

- **Intuition:** large numerator = projected means far apart; small denominator = projected clouds tight. We want big separation relative to within-group spread.

6.5 Solution via the Maximisation Lemma

- Re-parametrise: $\underline{u} = \Sigma^{1/2}\underline{a}$, so $\underline{a} = \Sigma^{-1/2}\underline{u}$. The ratio becomes:

$$\frac{\underline{u}^T \Sigma^{-1/2} B \Sigma^{-1/2} \underline{u}}{\underline{u}^T \underline{u}}$$

This is a **Rayleigh quotient** – the same object maximised in PCA.

- Spectral decomposition (Maximisation Lemma):

$$\Sigma^{-1/2} B \Sigma^{-1/2} = \sum_{i=1}^s \lambda_i \underline{e}_i \underline{e}_i^T, \quad s = \min\{p, g-1\}$$

- **Why $s = \min\{p, g-1\}$?** B has rank at most $g-1$ because it is built from g mean vectors subject to one linear constraint (they must average to $\bar{\mu}$).
- The optimum is $\underline{u}^* = \underline{e}_1$ (eigenvector for largest eigenvalue λ_1), giving:

$$\underline{a}_1 = \Sigma^{-1/2} \underline{e}_1, \quad \underline{a}_2 = \Sigma^{-1/2} \underline{e}_2, \quad \dots, \quad \underline{a}_s = \Sigma^{-1/2} \underline{e}_s$$

6.6 Fisher's Discriminant Scores

- The projections $\underline{a}_1^T \underline{X}$, $\underline{a}_2^T \underline{X}$, $\dots$ are called **First, Second, ... Fisher's Discriminant Scores**.
- Key property – scores are **uncorrelated**:

$$\text{Cov}(\underline{a}_i \underline{X}, \underline{a}_j \underline{X}) = \underline{a}_i^T \Sigma \underline{a}_j = \underline{e}_i^T \Sigma^{-1/2} \Sigma \Sigma^{-1/2} \underline{e}_j = \underline{e}_i^T \underline{e}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \implies \text{Cov}(A \underline{X}) = I$$

- In practice estimate with training data: $\hat{\underline{\mu}}_i = \bar{\underline{X}}_i$ and

$$\hat{\Sigma} = S_{\text{pooled}} = \frac{1}{n-g} \sum_{i=1}^g (n_i - 1) S_i$$

6.7 Classification via Voronoi Tessellation

- Project every observation and every group mean onto the first k Fisher directions (cut at level $k \leq s$ as in PCA).
- Assign a new point $\underline{X}$ to the group whose projected mean is nearest:

$$\delta(\underline{x}) = i \iff \sum_{j=1}^k (\underline{a}_j^T \underline{x} - \underline{a}_j^T \bar{\underline{X}}_i)^2 \leq \sum_{j=1}^k (\underline{a}_j^T \underline{x} - \underline{a}_j^T \bar{\underline{X}}_h)^2 \quad \forall h \neq i$$

- The resulting partition of the Fisher-coordinate plane is a **Voronoi tessellation**; boundaries between regions are **linear** (perpendicular bisectors of pairs of projected means) because we minimise squared Euclidean distance.
- **Theorem:** this classifier equals **LDA** when priors are equal: $p_1 = \dots = p_g = 1/g$.
- **Note:** $k = 2$ or 3 is used for visualisation; for actual classification use full LDA.

Summary Table

Topic	What it does	Why it matters
Naive Bayes	QDA with diagonal Σ_k	Works when $p > n$
AER	True misclassification probability	Gold standard; not computable
APER	Error on training set	Easy but optimistically biased
LOO	Test each unit once	Honest; high variance
k -fold CV	Split into k parts	Speed–variance compromise
Precision	$TP / (TP + FP)$	Costly false alarms
Recall	$TP / (TP + FN)$	Costly missed positives
F-measure	Harmonic mean of P & R	Balanced single metric
Fisher LDA	Maximise between/within ratio	LDA without Gaussianity; supervised dim. reduction

Türkçe Özet Notlar

1. Büyük Resim

- **Amaç:** Elimizde p özellikli veri noktaları var ve bunları g gruptan birine atamak istiyoruz.

$$\delta : \mathbb{R}^p \rightarrow \{1, 2, \dots, g\}$$

- Bu derste üç soru ele alınıyor: (1) kovaryans ters çevirilemiyorsa? $\rightarrow$ Naive Bayes. (2) Sınıflandırıcı ne kadar iyi? $\rightarrow$ AER, APER, çapraz doğrulama. (3) Gaussian varsaymadan LDA türetilbilir mi? $\rightarrow$ Fisher.

2. Naive Bayes Sınıflandırıcısı

- **Problem:** $p > n_k - 1$ ise S_k tekil (ters çevrilemez).
- **Çözüm:** Özellikler grup içinde **bağımsız** sayılır $\rightarrow \Sigma_k$ köşegen matris olur $\rightarrow$ kolayca ters çevrilir.
- k grubu için diskriminant fonksiyonu:

$$d_k^Q(\underline{x}) = \log(p_k) - \frac{1}{2} \log(\det \Sigma_k) - \frac{1}{2} \sum_{j=1}^p \frac{(x_j - \mu_{kj})^2}{\sigma_{jj}^{(k)}}$$

- Özünde: “Bu nokta hangi grubun ortalamasına *en yakın*?” sorusunu her özelliği kendi yayılımıyla ölçekleyerek soruyoruz.
- **Neden “Naive”?** Bağımsızlık varsayımı genellikle yanlış ama sıralama doğru olduğu sürece sınıflandırma çalışıyor.
- **Uyarı:** Önce PCA, sonra sınıflandırma **kötü bir yöntemdir**. PCA toplam varyansı maksimize eder; Fisher ise grupları ayıran yönü bulur.

3. Sınıflandırıcıyı Değerlendirme

- **AER (Gerçek Hata Oranı):** Nüfustan rastgele alınan bir birimin yanlış sınıflandırılma olasılığı. Hesaplanamaz, tahmin edilmek zorunda.

$$AER(\delta) = \sum_k \sum_{i \neq k} \int_{R_k} f_i(\underline{x}) p_i d\underline{x}$$

- **APER (Görünür Hata Oranı):** Eğitim setindeki hata oranı.

$$APER(\delta) = \frac{n_{12} + n_{21}}{n}$$

- **Neden kötü?** Sınıflandırıcıyı öğrendiğin veriyle test ediyorsun $\rightarrow$ daima iyimser çıkar.

4. Çapraz Doğrulama

- **Temel fikir:** Eğitim ve test verisi **ayrı** olmalı. Elimizde tek veri seti varsa simüle ederiz.
- **LOO (Birir Birir Çıkar):** Her birimi sırayla çıkar, geri kalanla eğit, çıkarılanı test et.

$$\widehat{AER}_{LOO} = \frac{1}{n} \sum_{i=1}^n \epsilon_i$$

Dürüst ama **yüksek varyanslı** – çünkü n sınıflandırıcı neredeyse aynı veriden öğreniliyor, hataları güçlü korelasyonlu.

- **k -fold:** Veriyi karıştır, k parçaya böl, her parçayı sırayla test et, ortalamasını al.

$$\widehat{AER} = \frac{1}{n} \sum_{j=1}^k n_j Err_j$$

- **Neden k -fold daha iyi?** Parçalar daha farklı $\rightarrow$ sınıflandırıcılar daha az korelasyonlu $\rightarrow$ ortalama varyansı gerçekten düşürür.
- **Temel sezgi:** Ortalama, ancak bağımsız (ya da zayıf korelasyonlu) şeylerden alındığında varyansı azaltır.
- B kez tekrarlayarak güven aralığı hesaplanabilir – LOO’da bu **imkânsızdır**.

5. Karmaşıklık Matrisi Metrikleri

- **Precision (Kesinlik):** “Hasta dediğimin kaçısı gerçekten hasta?”

$$\text{Precision} = \frac{n_{11}}{n_{.1}}$$

- **Recall (Duyarlılık):** “Gerçek hastaların kaçını yakaladım?”

$$\text{Recall} = \frac{n_{11}}{n_{1.}}$$

- **Specificity (Özgüllük):** “Gerçek sağlıklıların kaçını doğru etiketledim?”

$$\text{Specificity} = \frac{n_{00}}{n_{0.}}$$

- **F-Measure:** Precision ve Recall’un harmonik ortalaması.

$$F = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Neden bu kadar metrik?** Doğruluk (accuracy) tek başına yanıltıcı. Örnek: 1000 kişiden 10’u hasta. “Kimse hasta değil” diyen model $\rightarrow$ %99 doğruluk ama Recall = 0.

6. Fisher'in LDA Argümanı

- **Amaç:** Gaussian varsaymadan LDA'yı türet $\rightarrow$ LDA Gaussian olmayan durumlarda da çalışır.
- **Fikir:** p boyutlu veriyi 1 boyuta projekte et. En iyi yön: gruplar arası yayılımın, grup içi yayılıma oranını maksimize eden yön.

$$\arg \min_{\underline{a}} \frac{\underline{a}^T B \underline{a}}{\underline{a}^T \Sigma \underline{a}}$$

- **Çözüm:** $\underline{a}_1 = \Sigma^{-1/2} \underline{e}_1$ – en büyük özdeğere karşılık gelen özvektör.
- En fazla $\min(p, g - 1)$ yön vardır (görselleştirme için $k = 2$ veya 3 kullanılır).
- **Voronoi Tessellation:** Fisher koordinatlarında her noktayı en yakın grup ortalamaasına ata $\rightarrow$ sınırlar doğrusal.
- **Teorem:** Eşit prior ile Fisher sınıflandırıcısı = LDA.